Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Matthias Fahland	Email	Matthias.fahland@fep.fraunhofer.de
Institute Name	Fraunhofer Institute for Electron Beam and Plasma Technology (FEP)			
Address	Winterbergstrasse 28, 01277 Dresden, Germany			

2. Technology Overview

Title of Proposed Technology	A technology that allows maintaining temperature fluctuations within a range of ±3℃.			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility (x), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (x), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Chemicals		Electrical / Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Polymeric materials	Nanomaterial technology	Measuring instruments	Measuring sensors and components
Concept of Proposed Technology	The final technological goal is the development of a functionalized transparent window film with both heat blocking and anti-pathogenic functions The development program will be carried out on pilot-scale roll coating machines. This paves the way for an easy take-up of the technology towards an industrial production after the end of the project. This should preferably be done in a specially designed scale-up project. The concept includes various key technologies: <u>List of key technologies</u> 1. Development of Composite High-Function Nanomaterials • Thermal barrier metal/metal oxide composite heat shield design • Optimization of photocatalyst composition based on g-C_3N_4 - Simulation based design between transparency, durability and absorption characteristics 2) Development of Multi-layer Nano Thin Film Coating Technology			

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	• Oxide-Metal-Oxide (OMO) Composite Structure Film Design • Development of near-infrared reflection/absorption layer structure; Implementation of the multilayer- composite nano-coating process into an existing Roll-to-roll pilot coater • Introducing Flash crystallization technology - Enhanced oxidation resistance with interlayer ceramic interfacial layer coating 3) Photocatalytic sterilization function rear coating • Combining the technology with PVD coatings for accelerating Visible Light Reaction. • High density sputtering coating adjacent to the g-C_3N_4-coating; • Verification of photocatalytic performance 4) Product design and lamination/adhesion process development • Leveling Hard Coating • Application of PSA or other adhesive Layer • Structural verification of finished products using PET 5) Development of in-situ monitoring system • Optical and electrical sensing data collection - Machine learning-based outlier detection and automatic correction; process and quality optimization. • Classifying failure detection These key technologies shall help to achieve the following goals. 1) Solving current market issues; building high reliability / improving durability characteristics; applying anti-oxidation nano-coating crystallization flash process to realize durability of thermal barrier properties 2) Improvement of indoor air quality - Anti-pathogenic coating based on semiconductor photocatalyst g-C_3N_4-properties, prepared by sputter deposition The concept of the proposed project includes the development and specification of large area R2R vacuum PVD concept. This can be a basis for the concept of a future roll-to-roll production equipment. It includes: • Providing the specification of gas management • Providing the specification of heat load management • Providing a draft layout of the coater
Rationale for Joint R&D Project	Most of the domestic Korean automobile and construction heat shielding film companies • officially import and sell products from global companies. • or, it is made up of Toray Advanced Materials or Asia-Pacific/Korea branches of global companies such as Rintec Korea. In the case of domestic companies, SKC and Hyundai Motor are developing similar products, but they follow the level of heat blocking. The differentiated market is not expanding. In other words, R&D is needed for broad application fields and new product development, but due to intensifying competition by product, technology transfer is not possible due to national technology protection policies, and international technology competition is intensifying. The TIPA project provides an excellent opportunity to tackle the abovementioned issues by combining the German and Korean competencies. 1. It is necessary to continuously respond to new products based on the characteristics of consumer trend products, but it can be responded only when additional development/vacuum plasma WEB platform is established on the basis of the basic manufacturing process. 2. One of the reasons why it is difficult for domestic companies to establish a process technology foundation is to secure a nano-coating lamination process foundation for vacuum coating technology, and it is essential to advance process technology in an environment

	where a vacuum-based coating process is established. 3. Germany's Fraunhofer has process technology for this core process part, and as a joint development, it naturally intends to transfer core technologies and build localization of domestic technologies. The project aims to solve the current problems of the market. There is no production available that meets the requirements for broad market acceptance. The coated material includes an improvement of the anti-oxidation structure with enhanced durability, and develop a transparent nano-coated FILM manufacturing platform technology that simultaneously implements transparent near-infrared (NIR) heat blocking and anti-pathogenic complex photocatalyst functions by solving the market's current problems and optimizing AI design simulation. This technology is 1. It is a continuous nano-coating manufacturing composite technology consisting of multi-layered thermal barrier thin film composite design technology, 2. reactive g-C_3N_4 based composite photocatalytic coating with indoor air purification function, 3. the perspective to a 1.6 m roll-to-roll (R2R) vacuum composite PVD coating process and system construction, 4. silica leveling hard coating and adhesive layer laminating process. <u>Differentiation</u> Unlike conventional commercial film technology, which focuses on a single function (heat blocking function), this task is a composite of organic and inorganic hybrid structures Combining nano-thin film design with graphic C_3N_4 semiconductor, It provides both indoor air purification and heat blocking functions that decompose pollutants only with visible light such as sunlight and indoor light. Thus, it is a next generation film material with big market potential. Eco-friendly construction, new materials, and next-generation automobiles autonomously by establishing a large-area production technology with a width of 1.6 m required by the market compared to existing 60 cm experimental equipment It secures a source process platform that can be expanded to large-area construction fields such as driving, hospitals, and postpartum care centers.
Key Words	1. Roll-to-Roll coating processes 2. Sputter technology 3. Hard coating by slot die coating and electron beam curing 4. Semiconductor photocatalyst 5. Heat blocking coatings

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (6)
R&D Objectives	The key R&D objectives are Technology Development Goals • Development of a functionalized transparent window film with both heat blocking and anti-pathogenic functions • Development of a manufacturing technology for the functionalizes window film on pilot level • Demonstration of the performance of the functionalized window film including customer acceptance tests • Preparation of establishing production capabilities for the functionalized window film The key R&D objectives are motivated by the following considerations: 1. <u>The need to create industrial competition</u> There is an increasing global competition and lack of domestic Korean technological foundation in this field. Currently, products from global companies such as 3M, Saint-Gobain, Eastman (LLumar), Toray, and Mitsubishi Chemical account for more than 70% of the domestic automobile and construction heat blocking film market. Domestic companies are mainly importing and constructing products from overseas manufacturers, and the technology base for independently manufacturing high-function nanotechnology-based heat blocking and antibacterial composite films is very limited. In particular, vacuum plasma nanocomposite coating (Web coating) technology is a high-level technology monopolized by research institutes in some advanced countries such as Germany's Fraunhofer FEP, and localization of key element technologies such as large-area R2R complex process, multi-layer nano-lamination, and plasma simulation technology is insufficient in Korea 2. <u>The need for composite functional materials in response to the increasing demand for quarantine and energy saving</u> Since the post-COVID-19 era, the demand for air quality management in hospitals, nursing homes, and children's facilities has increased, and at the same time, the demand for cooling energy reduction in buildings due to climate change is rapidly increasing However, there are currently no transparent film products equipped with both "heat blocking functions" and "antibacterial and deodorizing functions" in the market, and the demand for composite functional films is increasing rapidly in high value-added markets such as public procurement, smart buildings, and autonomous vehicles. 3. <u>The need to preoccupy domestic technology and replace imports</u> This technology is a convergence technology that integrates photocatalyst-based anti-pathogenic functions, NIR heat blocking functions, and large-area vacuum R2R processes, enabling import substitution effects and securing domestic manufacturing bases when entering the market.			
R&D Target Specifications	The aim of the project is the development of a functional film with distinct properties, The specifications imply the properties themselves, but also the technological steps in the anticipated readiness level. <u>Summary of film properties</u> • Reduction of the transmitted radiation energy in the NIR spectral range to no more than 25% of the clear glass reference value • Keeping the visual transmittance of the film at ≥80% of the clear glass reference value • Determining adapted and relevant test procedure and metrics for antipathogenic properties			

	• Outdoor stability testing over 1 year (performed in test site on the roof of Fraunhofer FEP) Technological Readiness • All technologies are presented at roll-to-roll pilot scale level on a width of 600 mm in a continuous process. • The content includes an outlook to industrial-relevant width of 1.6 m Further topics. • Determining adapted and relevant test procedure and metrics for antipathogenic properties • The film is ready-to apply (i.e. it contains an adhesive with allows a retrofit of the possibility for the implementation into a production flow.
R&D Plan	The R&D plan can be divided in the following work packages. The detailed work plan can be adapted a fine- tuned in the cooperation with the entire consortium. This fine -tuning includes in particular the timing of the work packaged and the completed GANTT chart WP1: refining of the product specification (to be started at the beginning of the project) Content of WP 1: • Determining the specification of the developed film • Survey of substrate films • Determination of the layer structure • Determining of performance and specification of each layer stack component in the view of the complete performance requirements of the film Expected outcome of WP1 • Full specification of the anticipated properties • Detailed description of test procedures WP2: Material development (to be started after the completion of WP1) Content of WP2 • Development of heat blocking layer stack. Important parameters like plasma frequency (for heat blocking layers) or porosity (for barrier and protective layers) are demonstrated on a polymer film, • Development of anti-pathogenic layer (including structural analysis and functional tests according to the procedure determined in WP1), • Development of hard coat and adhesive layer; performance test according to the specifications fixed in WP1. Expected outcome of WP2 • All single materials are qualified • Test procedures for the single materials are established and provided the expected results WP 3: Technology development (WP3 will be performed in parallel to WP2) Content of WP3 • Roll-to-roll plasma-processes, preferably based on plasma processes. • Wet coating and electron beam curing for the hard coating • Wet coating and/or lamination for the adhesive component • Smart inline monitoring systems based on contemporary AI- and machine learning principles

are anticipated.

Expected outcomes of WP3:

- All processes ready for system integration

WP4 Layer stack system integration (to be started after the completion of WP2 and WP3):

Content of WP4

- All single layers and technologies are combined
- The final layer structure is demonstrated on the target substrate
- Performing test procedures according to the outcome of WP1

Expected outcomes of WP4:

- The compatibility for all components of the final system is demonstrated
- The performance of the complete system is demonstrated

WP5 Pilot demonstration (to be started after the completion of WP4):

Content of WP5

- Long term runs on the pilot line
- Providing test material for the preparation of commercial realization
- Establishing the inline monitoring system including contemporary AI and machine learning tools
- Test procedures according to the outcomes of WP1

Expected outcomes of WP5:

- The functionality of the technology is demonstrated on PILOT-level
- The stability of the performance of the coated material is demonstrated
- The inline monitoring system is ready for a transfer to the commercialization. The trained AI and machine learning solution supports the transfer of pre-knowledge

WP6: Preparation of industrial realization (to be started after the completion of WP2 and WP3, extending to the end of the project):

Content of WP6

- Study for the technical upgrade from the PILOT line to a 1.6 m wide production line
- Life cycle analysis (LCA)
- Cost analysis

Expected outcomes of WP6:

- Profound study including all the abovementioned topics as
  - a basis for the business plan for the commercialization
  - a support for the decision about investment in a production plant,

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute will actively participate in all the work packages outlined in the section before The research institute will take the leadership in the following work packages: • WP2 (material development) • WP3 (technological development) • WP4 (system integration) • WP5 (pilot demonstration) The research institute will determine the technological approach for the following components and technologies: • Plasma coating for the heat blocking layers • Wet coating and electron beam curing for the hard coating • LCA on the technology level The research institute will realise the technological approach of all components in experimental (WP2-WP4) and pilot levels (WP5)
Expected R&R of Korean SME Partner	The Korean SME will actively participate in all the work packages outlined in the section before The research institute will take the leadership in the following work packages: • WP1 (material development) • WP6 (Preparation of industrial realization) The Korean SME will determine the technological approach for the following components and technologies: • Plasma coating for the antipathogenic layers • Wet coating for the adhesive coating • Basic structure of AI and machine learning algorithms. (This is an offer to ease the later transfer to commercialization. FEP is ready to take leadership as well if this is required by the SME) • LCA on product level • Cost analysis The Korean SME will support the technological approach of all components in experimental (WP2-4) and pilot levels (WP5). The support should primarily include the following activities • Performing relevant tests (according to the outcome of WP1) • Performing customer acceptance studies, if needed

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	This study can grow into a strategic technology that leads the global market by establishing a platform technology for manufacturing high-function composite nano films that is absent in Korea. Transparent films that combine heat blocking and anti-pathogenic functions are essential core materials in all areas of architecture, automobiles, and medical care, and are highly market-expandable. It is expected to dramatically strengthen the competitiveness of the domestic material industry through process localization. (1) expected effect for technology • Localization of composite nanofilms that provide transparent heat blocking + anti-pathogenic functions at the same time • Securing superior thermal barrier performance (TSER, TSET) compared to existing metal oxide-based multi-layers • Implementation of anti-pathogenic air purification and organic decomposition deodorization functions through C_3N_4 semiconductor photocatalyst operated by visible light/indoor bulb • Evaluation of economic feasibility towards a 1.6 m wide roll-to-roll vacuum process (based on an operating 0.6m wide PILOT plant) and securing PILOT manufacturing base. • Process data based AI in-situ monitoring quality management module consumer customized system. (2) Economic and Industrial Expected Effects • Developed products are used as key components connecting the material industry (rear) – building, automotive, and medical markets (forward) • Replacing imports and improving trade balance through localization of premium heat shield film market (domestic 850 billion, global $5.7 billion) • Creation of new markets by expanding new applications such as construction, autonomous driving, hospitals, and nursing facilities • PILOT – Expectations of job creation and regional economic ripple effects from the conversion to production (20+) (3) Expected social effect • Effects of improving air quality and preventing infectious diseases in hospitals, nursing homes, and residential facilities • Reduce cooling and heating costs and carbon emissions through the application of high-efficiency energy-saving films • Reduce social costs only by film construction without replacing double-decker glass in existing old houses • Application of Smart WINDOW for eco-friendly electric vehicles (External heat blocking function / Internal air purification function simultaneously provided. Organic matter decomposition, etc.) Contributes to reducing environmental load

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6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization